Kabir Mathur

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Education

SVKM's Dwarkadas J. Sanghvi College of Engineering (DJSCE) 2023 – 2027
B.Tech in Computer Engineering (Data Science) · 8.648 CGPA · Honours in Financial Computing

Skills

Languages: Python, Java, SQL
AI/ML: Machine Learning, Deep Learning, NLP, Computer Vision, Reinforcement Learning, RAG, AI Agents, Agentic AI
Data Engineering: Apache Airflow, Apache Kafka, PySpark, PostgreSQL
Frameworks: TensorFlow, PyTorch, LangChain, LangGraph, n8n, Hugging Face, OpenCV, FAISS
Development & Cloud: Flask, Firebase, MongoDB, Docker, AWS

Experience

AI/ML Intern Sep. 2025 – Mar. 2026
Meera AI Tech Solutions LLP
* Built AI-powered Smart Glasses using YOLO, Qwen-2.5, and EasyOCR for real-time object detection, scene analysis, face recognition, and text reading to enhance accessibility for visually impaired users.
* Built a no-code AI platform for end-to-end mobile app development, generating full Android applications from prompts with an in-browser emulator for real-time preview and testing.

AI/ML Intern Jun. 2025 – Sep. 2025
K2S2 Digistrat Solutions
* Built AI-powered smart glasses for visually impaired users, combining YOLO (object detection), Qwen-2.5 (scene analysis), EasyOCR (text reading) and face recognition into a single real-time assistive pipeline.
* Built a no-code AI platform that turns a text prompt into a complete Android app, with an in-browser emulator so users can preview and test changes instantly without any setup.

Project Trainee Jul. 2026 – Present
Larsen & Toubro Precision Engineering and Systems
* Designed a gain-invariant, multi-channel time-frequency front end for CRNN acoustic classification, making the model robust to recording gain and microphone variation.
* Built and cross-validated a CRNN with attention pooling, replacing handcrafted features with learned representations and adding a harmonic-comb channel to reduce frequent misclassifications.

Projects

SafeGuard AI | GitHub | *YOLOv11, n8n, Flutter, Supabase, Three.js, Firebase* Mar. 2026
* Built an automated emergency response workflow using n8n that sends real-time alerts to hospitals, fire stations, and police stations from a single trigger, cutting manual dispatch time.
* Integrated a YOLOv11 model trained on 30,000 images to detect missing PPE equipment in real time, achieving an mAP score of 0.724 across 6 safety-gear classes.
* Developed a Flutter mobile app with Supabase authentication that receives live emergency alerts and shows per-worker PPE violation history and compliance trends.
* Designed a Three.js-powered site dashboard visualizing camera zones and live incident heatmaps synced with Firebase for real-time state updates.

Twinformer | GitHub | *PyTorch, Hugging Face Transformers, FAISS, FastAPI, React, TypeScript* Oct. 2026
* Built a Transformer latency Digital Twin (3 architectures) combining a physics-based simulator with a calibrated Ridge predictor, validated on a held-out sequence-length split.
* Engineered hybrid RAG (FAISS + BM25 RRF) over 9,113 chunks: 100% Hit-Rate@3, 0.71 MRR, 3.24 ms median retrieval.
* Developed an auto-routing chat system (5 task types) with attention-map extraction, KV-cache benchmarking, and a Groq-based LLM judge.
* Implemented scaling sweeps, a KV-cache reuse vs. recompute study, and a calibration ablation with an online loop that retrains the latency predictor after each chat.

Achievements & Certificates

* **Smart India Hackathon (SIH) 2024** – *Grand Finalist* **Dec. 2024**
* **Coders Arena 2025 – The Battle of Algorithms** – *Finalist* **July 2025**
* **Amazon ML Summer School** Certificate Link
* **AWS Machine Learning Certificate** Certificate Link
* **Generative AI for Data Science-Microsoft(Coursera)** Certificate Link
* **AWS Cloud Certificate** Certificate Link